

BIT Clubs & Communities

Discover • Connect • Participate

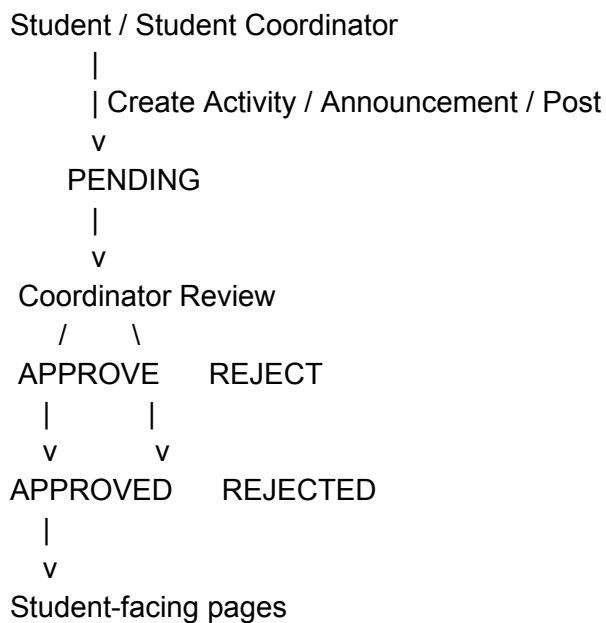
A full-stack web application for managing university clubs, memberships, activities, announcements, and club posts at Bannari Amman Institute of Technology (BIT).

1. Project Overview

BIT Clubs & Communities provides a centralized platform where students can discover clubs, join clubs, view activities and announcements, while coordinators can manage club information, members, activities, announcements and posts.

The application also supports a review/approval workflow for submitted content.

Core workflow



2. Main Features

Student

- Register and login
- Student dashboard
- Explore clubs
- Search/filter clubs

- View club details
- Join clubs
- View My Clubs
- View activities
- View announcements
- View approved club updates/posts

Coordinator / Administration

- Coordinator dashboard
- Pending approval list
- Approve/reject submitted content
- Manage clubs
- Add new clubs
- View club members
- View member counts
- View activity counts
- Activate/deactivate clubs
- Create activities
- Create announcements
- Create posts

The current coordinator UI is labelled **System Coordinator**.

3. Screenshots

Landing Page

<figure> <figcaption
aria-hidden="true">Landing Page</figcaption> </figure>

Login

<figure> <figcaption
aria-hidden="true">Login</figcaption> </figure>

Admin Dashboard

<figure>
<figcaption aria-hidden="true">Admin Dashboard</figcaption> </figure>

Create Activity

<figure> <figcaption
aria-hidden="true">Create Activity</figcaption> </figure>

Post Announcement

<figure> <figcaption aria-hidden="true">Post Announcement</figcaption> </figure>

Create Post

<figure> <figcaption aria-hidden="true">Create Post</figcaption> </figure>

Student Login

<figure> <figcaption aria-hidden="true">Student Login</figcaption> </figure>

Student Dashboard

<figure> <figcaption aria-hidden="true">Student Dashboard</figcaption> </figure>

Student Activities

<figure> <figcaption aria-hidden="true">Student Activities</figcaption> </figure>

Explore Clubs

<figure> <figcaption aria-hidden="true">Explore Clubs</figcaption> </figure>

Club Details

<figure> <figcaption aria-hidden="true">Club Details</figcaption> </figure>

Club Posts and Updates

<figure> <figcaption aria-hidden="true">Club Posts</figcaption> </figure>

Announcements

<figure> <figcaption aria-hidden="true">Announcements</figcaption> </figure>

Pending Approvals

<figure> <figcaption aria-hidden="true">Pending Approvals</figcaption> </figure>

No Pending Approvals

<figure> <figcaption aria-hidden="true">No Pending Approvals</figcaption> </figure>

Manage Clubs

<figure> <figcaption aria-hidden="true">Manage Clubs</figcaption> </figure>

Create New Club

<figure> <figcaption aria-hidden="true">Create New Club</figcaption> </figure>

Club Members

<figure> <figcaption aria-hidden="true">Club Members</figcaption> </figure>

My Clubs

<figure> <figcaption aria-hidden="true">My Clubs</figcaption> </figure>

Music Club Details

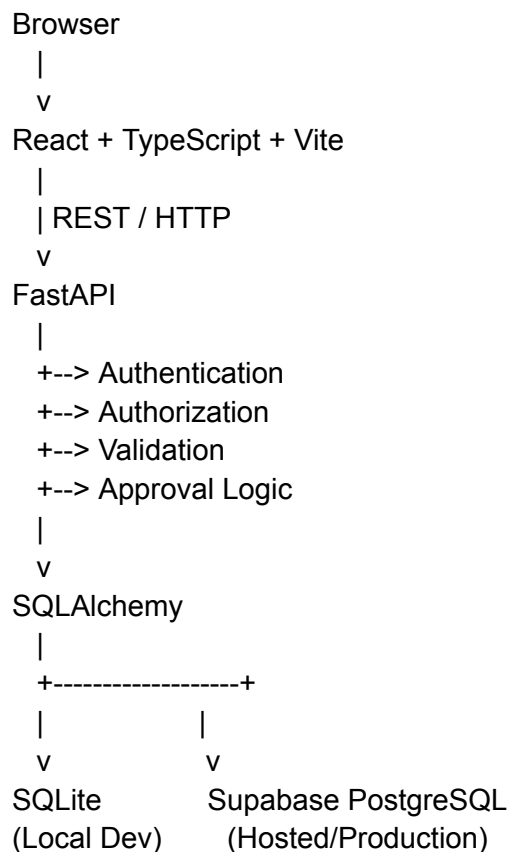
<figure> <figcaption aria-hidden="true">Music Club Details</figcaption> </figure>

4. Technology Stack

Layer	Technology	Purpose
Frontend	React + TypeScript	Component-based web UI
Build	Vite	Fast development/build tooling
Styling	CSS	Responsive application styling
Backend	FastAPI	REST API and business logic
Validation	Pydantic	Request/response validation
ORM	SQLAlchemy	Database access
Local DB	SQLite	Local development

Cloud DB	Supabase PostgreSQL	Hosted relational database
Authentication	Custom JWT	Existing application authentication
Optional Auth	Supabase JWT/JWKS	Optional migration path
Supabase	Python client	Supabase integration
Testing	Pytest	Backend testing
Server	Uvicorn	FastAPI server
Version Control	Git/GitHub	Collaboration and version control

5. Architecture



The frontend normally communicates with the backend API rather than connecting directly to the database.

6. User Flow

Student

Register / Login
 |
 v
 Dashboard
 |
 +--> Explore Clubs --> Club Details --> Join Club
 |
 +--> My Clubs
 |
 +--> Activities
 |
 +--> Announcements

Content submission and approval

Create Content
 |
 v
 PENDING
 |
 v
 Coordinator Review
 / \
 Approve Reject
 | |
 v v
 Approved Rejected
 |
 v
 Student-facing visibility

7. Supabase Integration

The project supports moving from local SQLite development to Supabase PostgreSQL.

Local development

FastAPI -> SQLAlchemy -> SQLite

Hosted database

FastAPI -> SQLAlchemy / Supabase Client -> Supabase PostgreSQL

The repository includes a migration helper for existing SQLite data.

Backend environment variables

SUPABASE_URL=<your-supabase-project-url>

```
SUPABASE_PUBLISHABLE_KEY=<your-publishable-key>
SUPABASE_SECRET_KEY=<your-secret-key>
SUPABASE_JWKS_URL=<your-jwks-url>
```

Frontend environment variables

```
VITE_SUPABASE_URL=<your-supabase-project-url>
VITE_SUPABASE_PUBLISHABLE_KEY=<your-publishable-key>
```

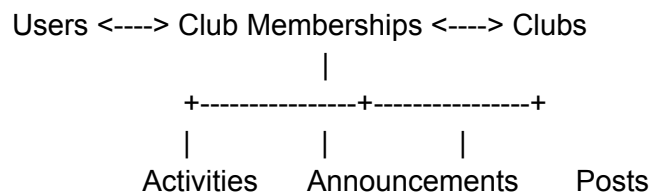
Important: never expose `SUPABASE_SECRET_KEY` in frontend code.

8. Database

Main application entities:

```
users
clubs
club_memberships
activities
announcements
posts
```

Relationship overview:



The relational structure supports member counts, club-based filtering, activity records and announcements.

9. Database Migration to Supabase

Before migration:

1. Create a Supabase project.
2. Open **Supabase Dashboard** → **SQL Editor**.
3. Execute the SQL from `database/schema.sql`.
4. Verify that the required tables exist.
5. Configure the backend `.env`.
6. Run:

```
python migrate_to_supabase.py
```

The migration helper is intended to upsert:

users
clubs
club_memberships
activities
announcements
posts

10. Authentication

The existing application uses custom JWT authentication.

Login

|

v

FastAPI validates credentials

|

v

JWT token

|

v

Authenticated API requests

|

v

FastAPI validates token

The project documentation also provides Supabase JWT/JWKS as an optional authentication migration path.

11. Authorization

Authentication identifies the user; authorization controls what that user can do.

Typical permissions are:

Student

- > Explore clubs
- > Join clubs
- > View student-facing content

Student / Club Coordinator

- > Submit/manage allowed club content

Coordinator / Administrator

- > Review pending submissions
- > Approve/reject
- > Manage clubs

-> View members

The final role-to-permission mapping should follow the current backend implementation.

12. API

FastAPI exposes API groups under `/api/....`

Typical groups include:

```
/api/auth  
/api/clubs  
/api/activities  
/api/announcements  
/api/posts  
/api/coordinator  
/api/admin
```

Swagger

`http://127.0.0.1:8000/docs`

ReDoc

`http://127.0.0.1:8000/redoc`

Swagger is useful for review because it exposes the API methods, request bodies, parameters, responses and authentication requirements.

13. Backend Setup

```
cd "d:/club management system/backend"
```

```
python -m venv venv
```

```
# Windows  
venv\Scripts\activate
```

```
pip install -r requirements.txt
```

If required:

```
pip install supabase PyJWT[crypto]
```

Configure `.env`, then start:

```
uvicorn app.main:app --host 127.0.0.1 --port 8000
```

14. Frontend Setup

```
cd "d:/club management system/frontend"
```

```
npm install  
npm run dev
```

Open:

<http://localhost:5173>

15. Complete Local Run

Terminal 1

```
cd backend  
venv\Scripts\activate  
uvicorn app.main:app --host 127.0.0.1 --port 8000
```

Terminal 2

```
cd frontend  
npm run dev
```

Then open:

<http://localhost:5173>

16. Project Structure

```
College-Club-Management-System/  
|  
+-- backend/  
|   +-- app/  
|   |   +-- main.py  
|   |   +-- config.py  
|   |   +-- supabase_client.py  
|   |   +-- ...  
|   +-- requirements.txt  
|   +-- .env  
|  
+-- frontend/  
|   +-- src/  
|   |   +-- components/  
|   |   +-- pages/  
|   |   +-- lib/  
|   |       +-- supabase.ts  
|   +-- package.json  
|   +-- .env
```

```
|  
+-- database/  
| +-- schema.sql  
|  
+-- migrate_to_supabase.py  
+-- README.md  
+-- LICENSE
```

17. Reviewer Demonstration

A clear review sequence is:

1. Open the landing page.
2. Login as a student.
3. Show the student dashboard.
4. Explore clubs.
5. Open a club.
6. Show My Clubs.
7. Show activities and announcements.
8. Login as coordinator.
9. Create an activity/announcement/post.
10. Show that the submission enters the pending workflow.
11. Open **Pending Approvals**.
12. Approve or reject the submission.
13. Return to the student-facing interface and demonstrate the resulting visibility.
14. Open **Manage Clubs**.
15. Create a club.
16. Open club members.
17. Show the API documentation at [/docs](#).

18. Testing Checklist

Authentication

- Registration works
- Login works
- Invalid login is rejected
- Protected routes require authentication
- Logout works

Student

- Dashboard loads
- Explore works
- Search/filter works

- Club details load
- Join club works
- My Clubs loads
- Activities load
- Announcements load

Coordinator

- Dashboard loads
- Pending approvals load
- Approve works
- Reject works
- Manage Clubs loads
- Add Club works
- Members page works
- Activate/deactivate works

Content

- Activity creation works
- Announcement creation works
- Post creation works
- Approval state is preserved
- Approved content appears where expected

Database

- SQLite works locally
- Supabase tables are created
- Migration script runs
- Supabase data can be read
- Secret keys are not exposed

19. Troubleshooting

Supabase table not found

If you see:

Could not find the table ... in the schema cache

Run the required SQL from:

database/schema.sql

in the Supabase SQL Editor and verify that the table exists.

CORS error

Ensure backend CORS configuration allows:

`http://localhost:5173`

Frontend Supabase variables not working

Vite variables must use the `VITE_` prefix:

`VITE_SUPABASE_URL=...`

`VITE_SUPABASE_PUBLISHABLE_KEY=...`

Restart Vite after changing `.env`.

Backend startup problem

```
python --version
```

```
venv\Scripts\activate
```

```
pip install -r requirements.txt
```

```
uvicorn app.main:app --host 127.0.0.1 --port 8000
```

20. Security

Never commit real secrets to GitHub.

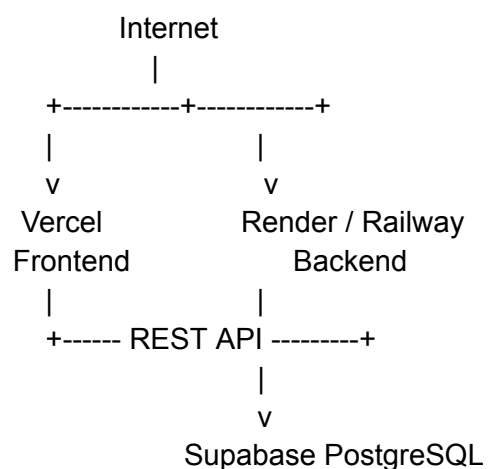
Do not expose:

`SUPABASE_SECRET_KEY`

in frontend code.

Use `.env.example` to document variable names without real credentials.

21. Deployment Architecture



The frontend can be built with:

npm run build

The FastAPI backend can be deployed behind an ASGI server.

22. Why This Architecture?

React

Used for reusable, component-based screens and dashboards.

TypeScript

Provides type safety for structured entities such as users, clubs, memberships and activities.

FastAPI

Provides the API boundary and centralizes authentication, authorization, validation and business logic.

SQLAlchemy

Provides structured database access and keeps database operations separate from UI code.

PostgreSQL / Supabase

Provides a hosted relational database suitable for users, clubs, memberships, activities, announcements and posts.

SQLite

Keeps local development simple without requiring a separate database server.

JWT

Provides stateless authentication for the existing application.

Supabase

Provides hosted PostgreSQL and supporting backend services for cloud deployment.

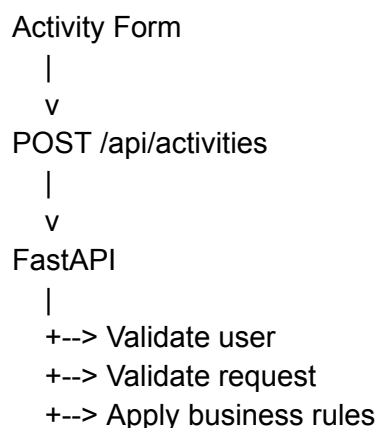
23. One-Minute Reviewer Explanation

BIT Clubs & Communities is a centralized web platform for managing college clubs and communities. Students can discover and join clubs, view activities and announcements, and manage their joined clubs. Coordinators can manage club members and create activities, announcements and posts. Submitted content can pass through a pending approval workflow where an authorized coordinator reviews it and approves or rejects it. The frontend uses React and TypeScript, the backend uses FastAPI with Pydantic and SQLAlchemy, and the database can run on SQLite during local development and Supabase PostgreSQL for hosted deployment.

24. Technical Request Flow



Example activity creation:



|
v
Database

Approval:

PENDING

|
+----> APPROVE ----> APPROVED
|
+----> REJECT -----> REJECTED

25. Screenshot Index

#	Screen	File
1	Landing Page	01-homepage.png
2	Login	02-login.png
3	Admin Dashboard	03-admin-dashboard.png
4	Create Activity	04-create-activity.png
5	Post Announcement	05-post-announcement.png
6	Create Post	06-create-post.png
7	Student Login	07-student-login.png
8	Student Dashboard	08-student-dashboard.png
9	Student Activities	09-student-activities.png
10	Explore Clubs	10-explore-clubs.png
11	Club Details	11-club-details.png
12	Club Posts	12-club-posts.png
13	Announcements	13-announcements.png
14	Pending Approvals	14-coordinator-pending-approvals.png
15	No Pending Approvals	15-coordinator-no-pending.png
16	Manage Clubs	16-manage-clubs.png

17	Create New Club	17-create-new-club.png
18	Club Members	18-club-members.png
19	My Clubs	19-my-clubs.png
20	Music Club Details	20-music-club-details.png

26. Final Summary

BIT Clubs & Communities

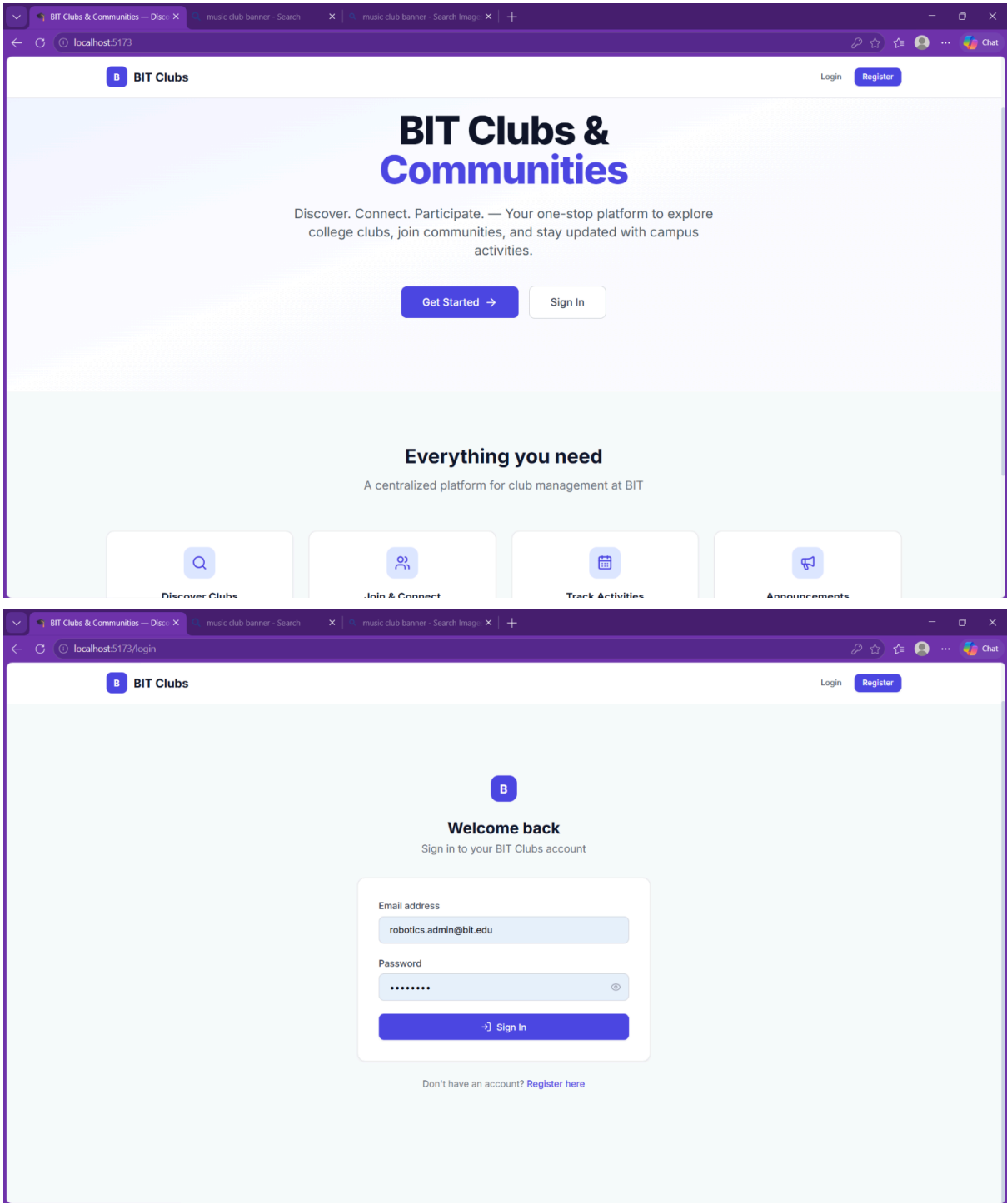
- |
- +--> Students
 - | +--> Discover clubs
 - | +--> Join clubs
 - | +--> View activities
 - | +--> View announcements
 - |
- +--> Coordinators
 - | +--> Manage club
 - | +--> Manage members
 - | +--> Create content
 - |
- +--> Approval
 - | +--> Pending
 - | +--> Approve
 - | +--> Reject
 - |
- +--> Backend
 - | +--> FastAPI
 - | +--> SQLAlchemy
 - |
- +--> Database
 - +--> SQLite
 - +--> Supabase PostgreSQL

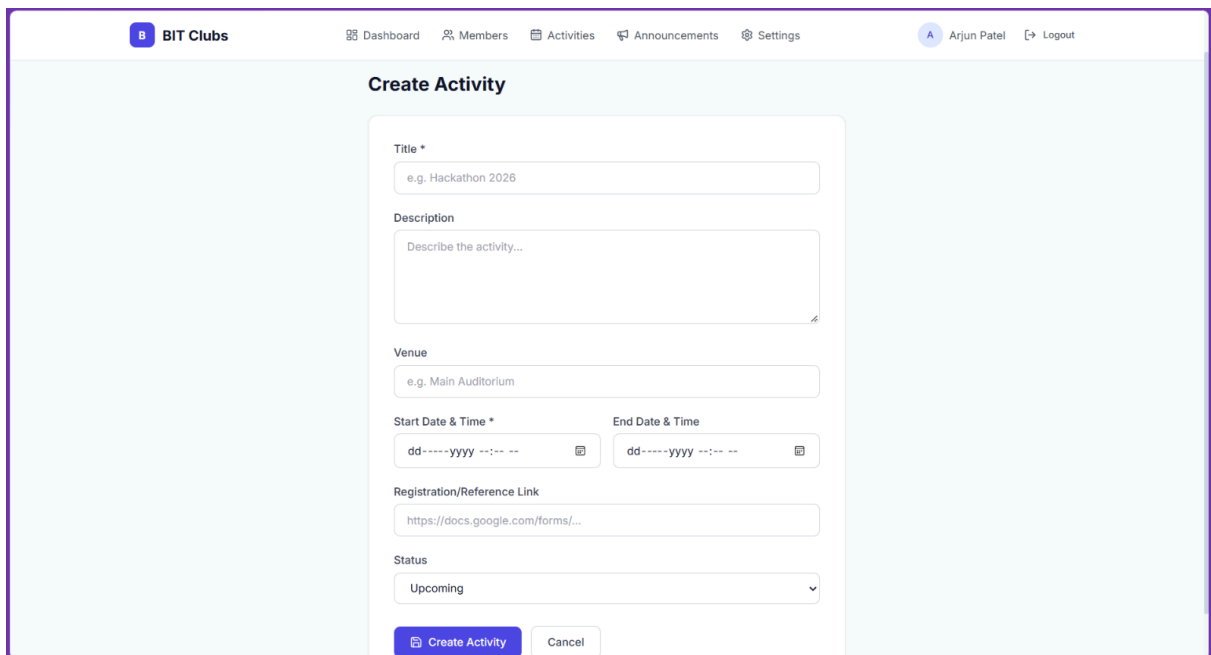
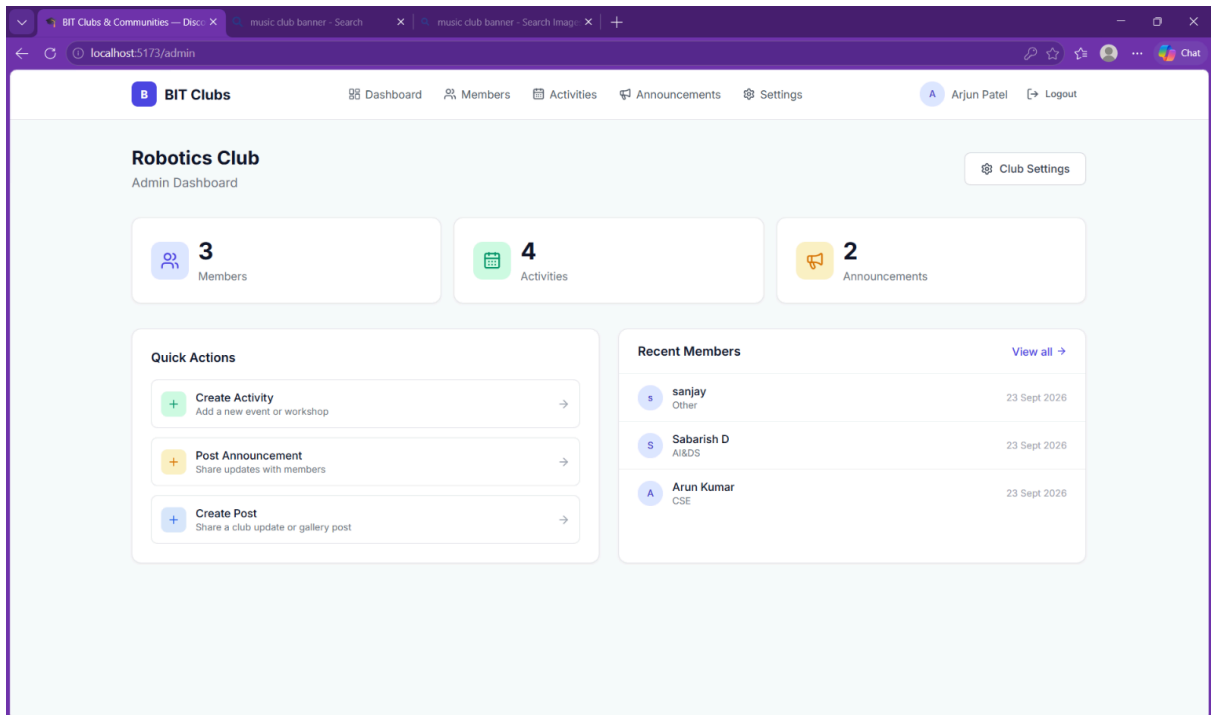
BIT Clubs & Communities

Bannari Amman Institute of Technology

Discover • Connect • Participate

HOME PAGE :





BIT Clubs

Dashboard

Members

Activities

Announcements

Settings

Arjun Patel

Logout

Back to Announcements

Post Announcement

Title *

e.g. Important: Registration Deadline

Content *

Write your announcement...

Reference Link

https://example.com/...

Priority

Normal

Post Announcement

Cancel

BIT Clubs & Communities — Disc

music club banner - Search

music club banner - Search image

localhost:5173/admin/posts/new

Arjun Patel

Logout

Back to Posts

Create Post

Publish a new update or gallery post on your club page

Posts require approval from the Club Coordinator before they are publicly visible on the club page.

Post Title *e.g., Highlights from our rec

Write your post content here...

Content *

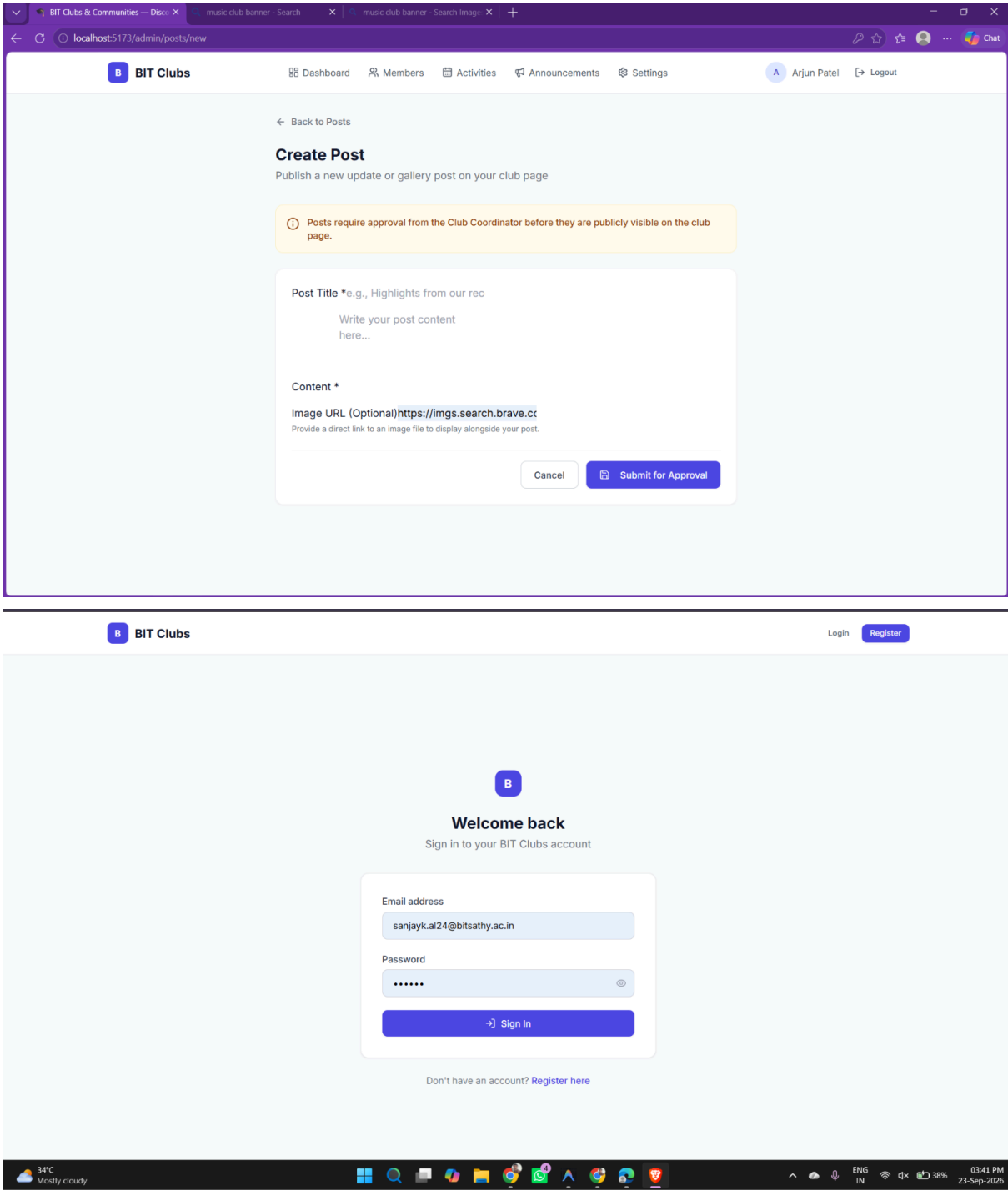
Image URL (Optional)https://imgs.search.brave.cc

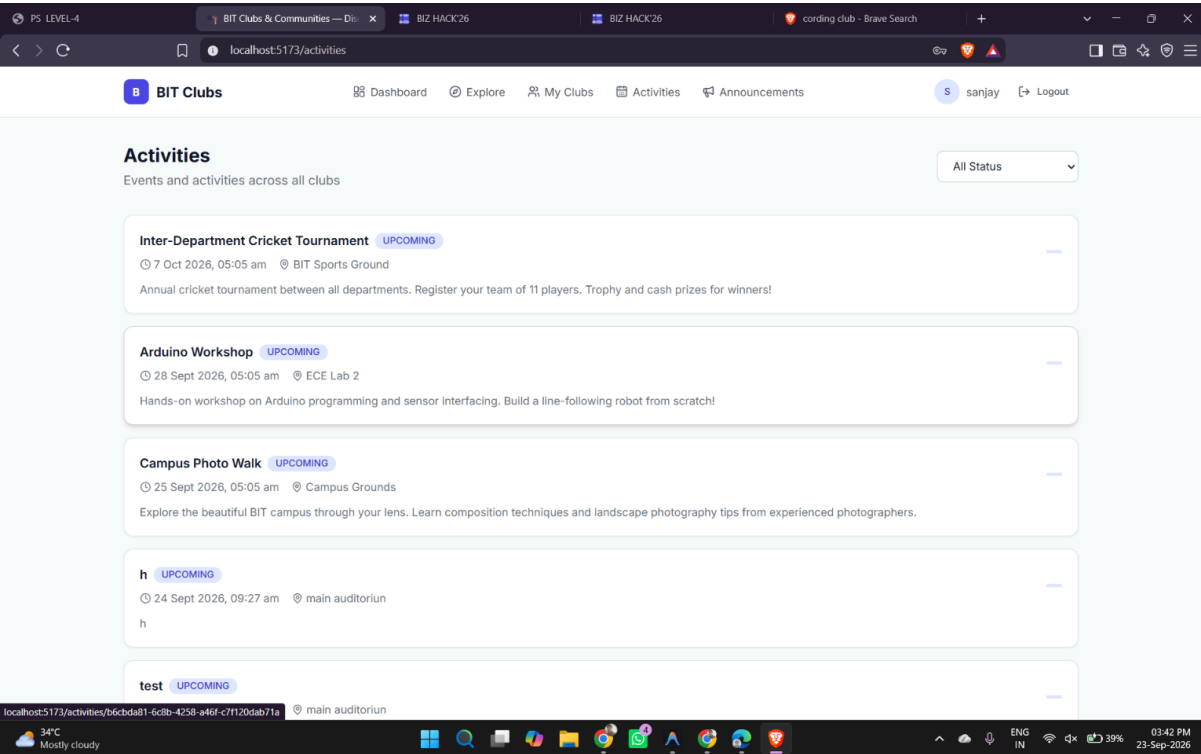
Provide a direct link to an image file to display alongside your post.

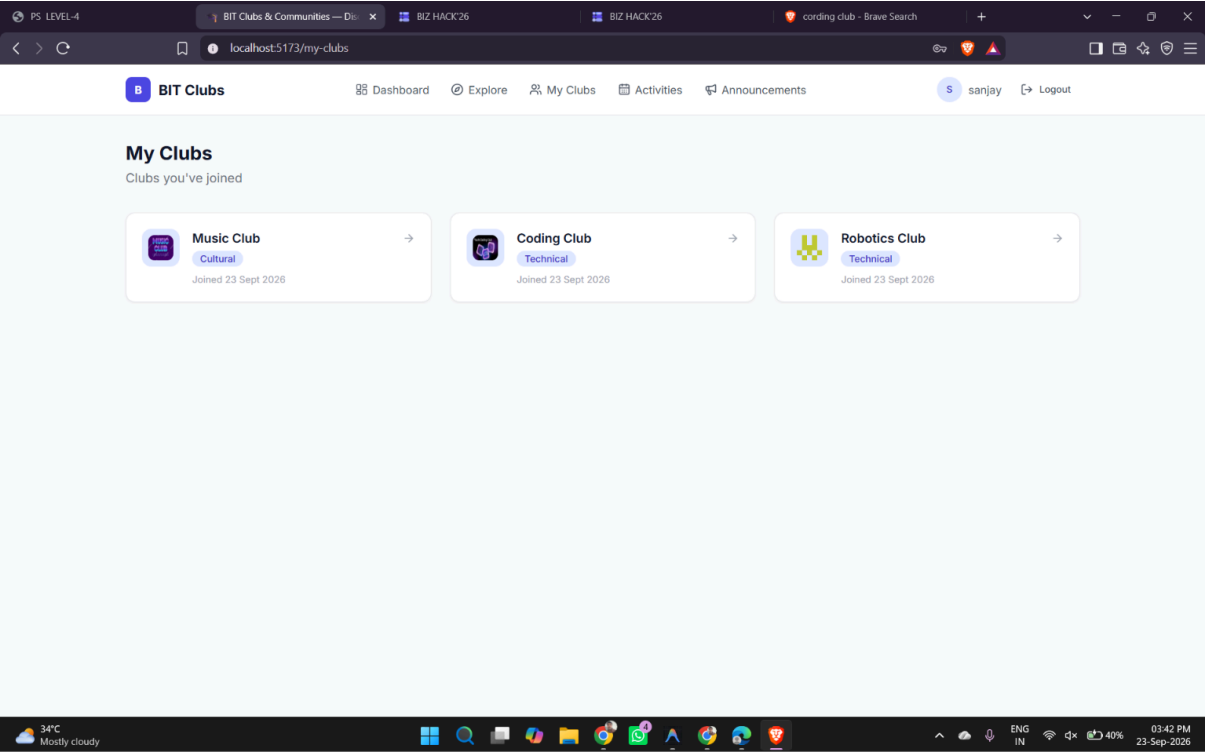
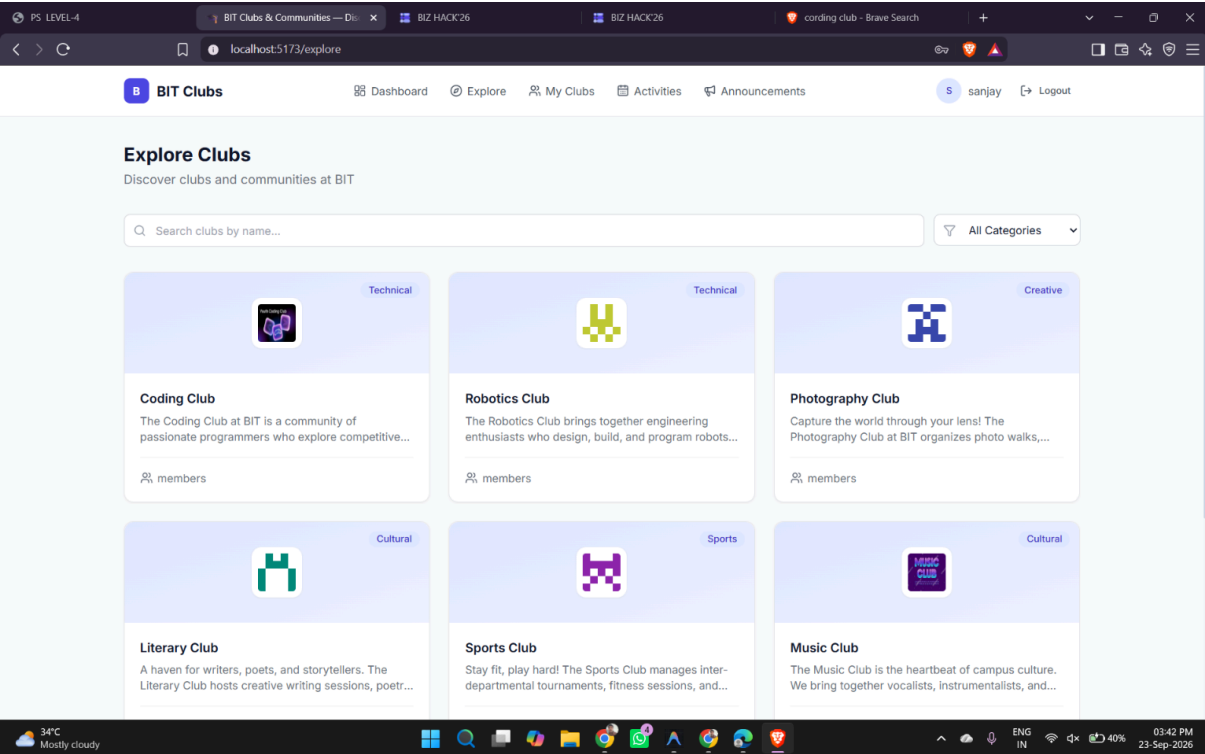
Cancel

Submit for Approval

STUDENT CODINATERPAGE:







PS LEVEL-4

BIT Clubs & Communities — D...

BIZ HACK'26

BIZ HACK'26

coding club - Brave Search

+

localhost:5173/clubs/music-club

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BIT Clubs

Dashboard Explore My Clubs Activities Announcements

sanjay Logout

← Back to Explore



Music Club

Cultural

members

Join Club

The Music Club is the heartbeat of campus culture. We bring together vocalists, instrumentalists, and music producers for jam sessions, concerts, and cultural fest performances. All genres and skill levels welcome — come make some noise!

Activities1

T

23 Sept 2026, 08:10 am @t

UPCOMING

Announcements1

Open Jam Session This Friday

Join us for an open jam session this Friday at 5 PM in the Music Room. Bring your instruments — amps and drums are available. All skill levels welcome!

NORMAL

Club Posts & Updates0

No posts yet

34°C Mostly cloudy



ENG IN 40% 03:42 PM 23-Sep-2026

PS LEVEL-4

BIT Clubs & Communities — D...

BIZ HACK'26

BIZ HACK'26

coding club - Brave Search

+

localhost:5173/clubs/coding-club

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BIT Clubs

Dashboard Explore My Clubs Activities Announcements

sanjay Logout

← Back to Explore



Coding Club

Technical

members

Join Club

The Coding Club at BIT is a community of passionate programmers who explore competitive programming, open-source contributions, hackathons, and emerging technologies. We host weekly coding challenges, tech talks, and collaborative projects to sharpen our skills and build innovative solutions together.

Activities4

test

24 Sept 2026, 08:53 am @ main auditorium

UPCOMING

san

24 Sept 2026, 08:30 am @ sa

UPCOMING

BIZ HACK'26 - One-Day Full Stack & Software Development Hackathon

23 Sept 2026, 05:44 am @ Learning Center

ONGOING

Test Activity Wikipedia Link

22 Sept 2026, 02:48 am

UPCOMING

Announcements4

Test Announcement URL

This is a test announcement to verify link functionality.

NORMAL

first year guid to choose the club

ww201 3:25 to 4:30

HIGH

Weekly Coding Challenge #15

This week's coding challenge is live on our platform. Solve the problem set and submit your solutions by Sunday 11:59 PM. Top 3 scorers get certificates!

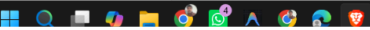
NORMAL

Hackathon Registration Open!

Registrations for Hackathon 2026 are now open. Form teams of 3-4 members and register before the deadline. Limited spots available!

HIGH

34°C Mostly cloudy



ENG IN 40% 03:43 PM 23-Sep-2026

Test Activity Wikipedia Link

22 Sept 2026, 02:48 am

UPCOMING

Hackathon Registration Open!

Registrations for Hackathon 2026 are now open. Form teams of 3-4 members and register before the deadline. Limited spots available!

HIGH

Club Posts & Updates

3

test

test



23 Sept 2026, 08:47 am

hai

PS LEVEL-4

BIT Clubs & Communities — Di

BIZ HACK26

BIZ HACK26

cording club - Brave Search

+

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localhost:5173/announcements

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☰

BIT Clubs

Dashboard

Explore

My Clubs

Activities

Announcements

S sanjay

Logout

Announcements

Latest updates from clubs

📢

Test Announcement URL

• 9h ago

NORMAL

This is a test announcement to verify link functionality.

📢

first year guid to choose the club

• 9h ago

HIGH

ww201 3:25 to 4:30

📢

Open Jam Session This Friday

• 10h ago

NORMAL

Join us for an open jam session this Friday at 5 PM in the Music Room. Bring your instruments — amps and drums are available. All skill levels welcome!

📢

Cricket Tournament Schedule Released

• 10h ago

HIGH

The schedule for the inter-department cricket tournament has been finalized. Check your match timings and ensure your team is ready. No rescheduling allowed.

📢

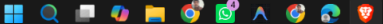
Photo Contest Results

• 10h ago

LOW

34°C

Mostly cloudy



ENG

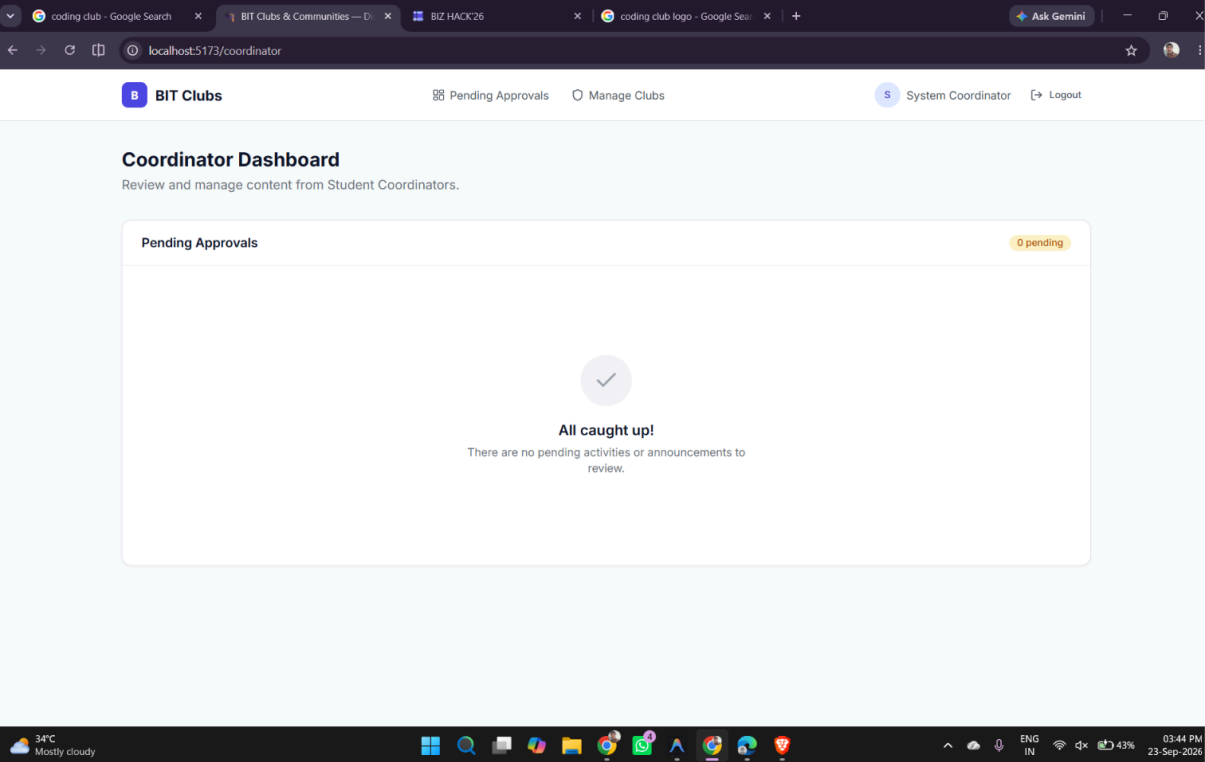
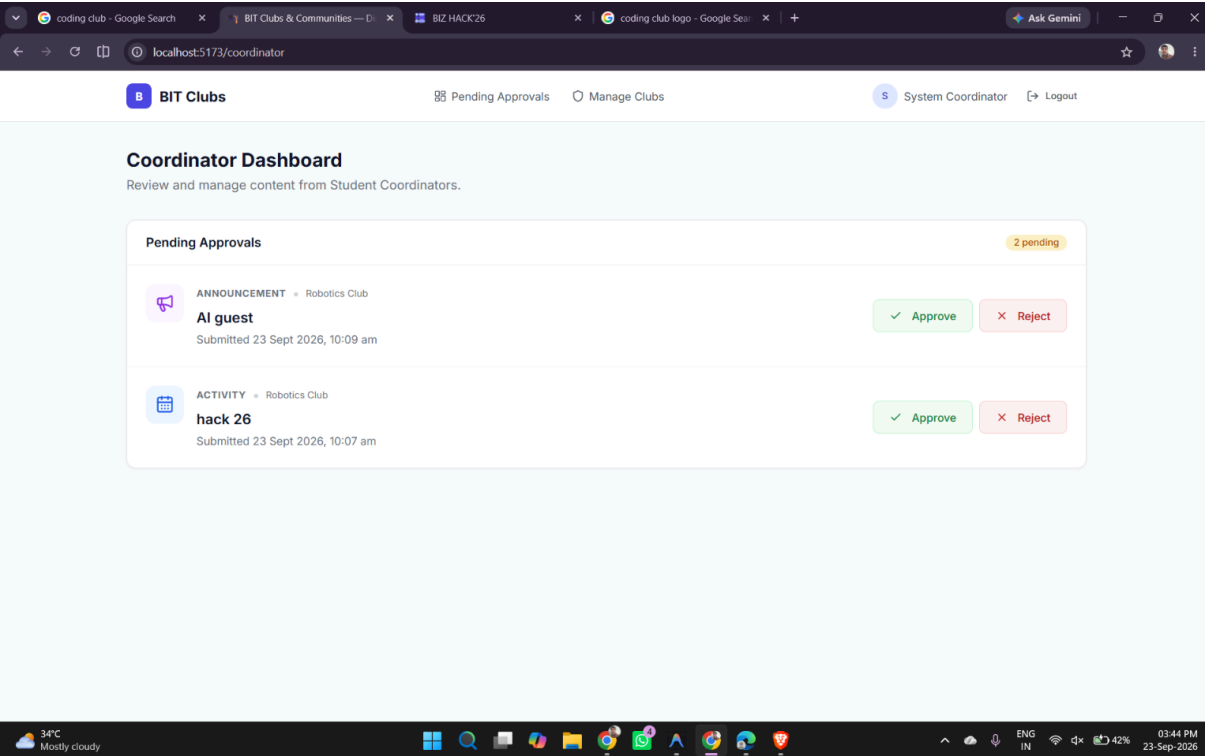
IN

📶

🔋 41%

03:43 PM

23-Sep-2026



View and manage all clubs under your supervision.

View and manage all clubs under your supervision.

+ Add New Club

A screenshot of the 'Photography Club' page in the app. The page has a white background with a light blue header bar. At the top right, there is a green badge with the word 'Active' in white. Below the header, the club name 'Photography Club' is displayed in a large, bold, black font, followed by the word 'CREATIVE' in a smaller, bold, black font. In the center, there are two light blue rectangular boxes. The left box contains a large blue number '1' above the word 'MEMBERS' in a bold, black font. The right box contains a large orange number '1' above the word 'ACTIVITIES' in a bold, black font. At the bottom of the screen, there are two white buttons with rounded corners. The left button has a blue person icon and the text 'Members' in a bold, black font. The right button has a red 'X' icon and the text 'Deactivate' in a bold, black font.

Music Club

CULTURAL

Active

4
MEMBERS

1
ACTIVITIES

 Members

 Deactivate

